

COMPUTER OVERVIEW

A **computer** can be defined as an electronic and electromechanical device, which receives the data from external/ internal sources, process that data and gives out the output in the form of processed data/information.

Data refers to business resources, raw facts, concepts, opinions or ideas that describe an organization i.e any material that constitutes a fact. It is treated as raw because unless or until the user receives the output from the system it does not hold any value for the user.

Information is derived from processing the data i.e. its first meaningful resources and therefore when processed manually, mechanically or electronically, information is obtained. Such form of processing task takes the form of classification, sorting, calculating, summarizing etc

Characteristics of a Computer

Speed

Computer is very fast machines that can process millions of instruction within a few seconds.

Accuracy

Computer gives consistent and accurate results. Computer results are accurate because it performs an action according to a given instruction.

Memory

Computers have enormous amount of memory to hold a very large amount of data. We can store large amount of data information in the secondary storage device.

No intelligence

Computers do not have any intelligence. They perform only those operations, which are already fed into them by the external sources.

Diligence

Computer is free from problems like lack of concentration, and confusion etc. It can work hours and hours without feeling tiredness.

Versatility/multitasking

We can perform many different types of tasks on a computer, one moment it might be busy in calculating the statistical data for annual performance evaluation of a business organization and next moment it might be working on weather forecast.

SELF STUDY.

Describe the application of computers in the society

Classification of computers

1. According to their functions.

- General purpose computers – that performs a variety of tasks
- Special purpose computers – that performs a particular task
- Dedicated purpose computers – that is a general purpose computer restricted to performing only a single task e.g. a server computer

2. According to how they process and receive data

- Digital computer – process data that is binary form presented as binary 1s and 0s.
- Analogue computer – process data in continuous form presented as physical quantity such as time, temperature and speed
- Hybrid computer – has features of both digital and analog computers

3. According to their size, cost, storage capabilities and processing power.

- Super computers
- Main frame computers
- Mini-computer/mid-range computers
- Micro-computer/desktop computers

Supercomputers

- They contain a number of processors, which operate in parallel to make them faster, giving them their speed through parallel processing.
- Capable of handling large amounts of data
- Uses multiprocessing and supports multiprogramming
- They are used for weather forecasting, weapons research and development, rocketing, aerodynamics, atomic nuclear and plasma physics. Supercomputers have limited use and limited market because of their very high price.

Mainframe computers

- Are very powerful, large-scale general-purpose computers.
- Mainframes are capable of processing data at very high speeds billions of instructions per second and have access to trillions of characters of data.
- They are more expensive compared to mini and micro computers
- They are used where large amount of data are to be processed or very complex calculations are to be made and these tasks are beyond the capacities of mini computers. They are also used as controlling nodes in WANs (Wide Area Networks) and used to manage large centralized databases

Minicomputers.

- Are small to medium scale computer that is the **midrange** between a microcomputer and a mainframe.
- They can support from a handful up to several hundred of user terminals.
- They have medium processing and storage capabilities
- They are used as servers in medium sized organizations
- They are extensively used for pay-roll preparation, accounting and scientific computation.
- These computers are used when the volume of processing is large for example data processing for a medium sized organization

Micro computers

- They are personal computers designed for an individual user.
- They are small in size.
- Personal computers are typically used for applications, such as word processing, spreadsheets, database management etc.
- They are the slowest in processing and storage capabilities
- Commonly used for domestic and office work.

4. According to Generations

In Computer language, “Generation” is a set of Technology. It provides a framework for the growth of the computer technology. There are totally Five Computer Generations till today. Discussed as following.

First Generation:

- Duration: 1942-1955
- Technology: vacuum tube
 - Used as a calculating device.
 - Performed calculations in milliseconds.
 - Too bulky in size & complex design.
 - Required large room to place it.
 - Generates too much heat & burnt.
 - Required continuously hardware maintenance.
 - Generates much heat so must air-conditioner rooms are required.
 - Commercial production is difficult & costly.
 - Difficult to configure.
 - Limited commercial use.
 - ENIAC, EDVAC, EDSAC are example of 1st generation computer.

Second Generation:

- Duration: 1955-1964
- Technology: transistor
 - 10 times Smaller in size than 1st generation system.
 - Less heat than 1st generation computers.
 - Consumed less power than 1st generation system.
 - Computers were done calculations in microseconds.
 - Air-conditioner is also required.
 - Easy to configure than 1st generation computers.
 - More reliable in information.
 - Wider commercial use.
 - Large & fast primary/secondary storage than 1st generation computers.

Third Generation:

- Duration: 1965-1975
- Technology: IC chip
 - Smaller in size than 1st & 2nd generation computers.
 - Perform more fast calculations than 2nd generation systems.
 - Large & fast primary/secondary storage than 2nd generation computers.
 - Air –conditioner is required.
 - Widely used for commercial applications.
 - General purpose computers.

- High level languages like COBOL & FORTRAN are allowed to write programs.
- Generate less heat & consumed less power than 2nd generation computer.

Fourth Generation:

- Duration: 1975-1989
- Technology: Microprocessor chip
 - Based on LSI & VLSI microprocessor chip.
 - Smaller in size.
 - Much faster than previous generations.
 - Minimum hardware maintenance is required.
 - Very reliable as computer to previous generation computers.
 - Totally general purpose computer.
 - Easy to configure.
 - Possible to use network concept to connect the computer together.
 - NO requirement of air-conditioners.
 - Cheapest in price.

Fifth Generation:

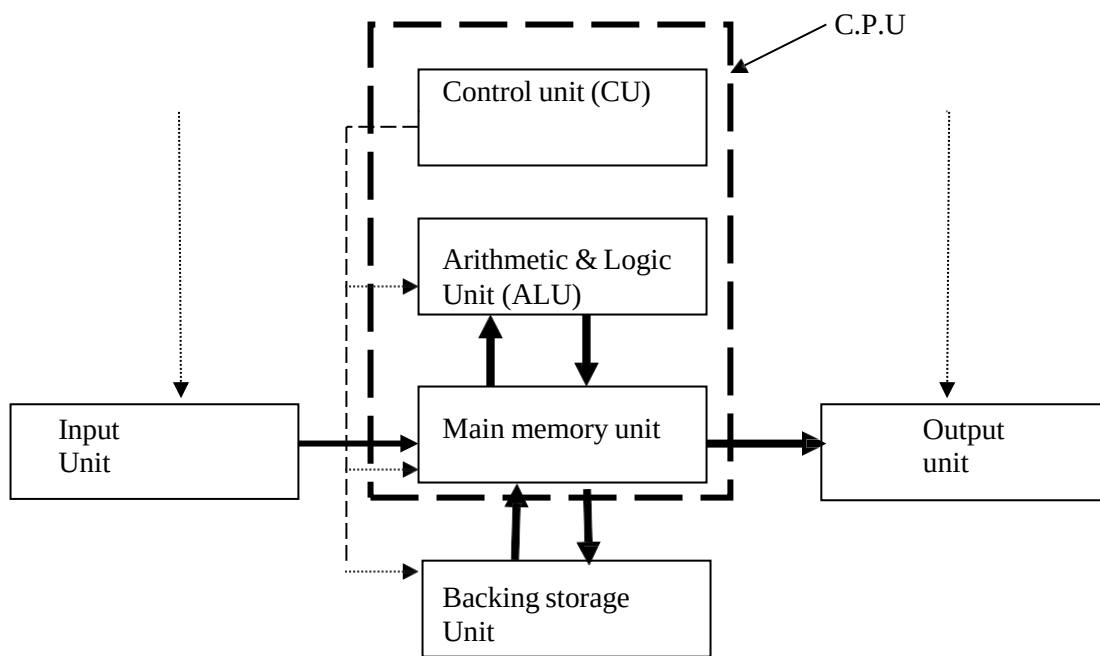
- Duration: 1989 to Present
- Technology: ULSI microprocessor chip
 - Much smaller & handy.
 - Based on the ULSI chip which contains 100 million electronic components.
 - The speed of the operations is increased.
 - Consumed less power.
 - Air-conditioner is not required.
 - More user friendly interface with multi-media features.
 - High level languages are allowed to write programs.
 - Larger & faster primary/secondary storage than previous generations.

FUNCTIONAL UNITS OF A COMPUTER

Computer is a device that operates upon information or data. It is an electronic device which accepts input data, stores the data, does arithmetic and logic operation and outputs the information in desired format.

Even though the size, shape, performance, reliability and cost of computers have been changing over the years, the basic **logical structure proposed by Von Neumann** has not changed. The internal architecture of computers differs from one system model to another.

A block diagram of the basic computer organization specifying different functional units is shown in the figure below.



Key:

.....> Command/instruction flow

————> Data/signal flow

CPU – Central processing unit

Input Unit

It supports the presentation of data to the computer. Input devices are devices that bring data into the computer. The purpose of input devices is to translate data and information from human-readable format to electrical impulses. Keyboard and Mouse are the examples of input devices.

Output unit

This is for provides information either in **soft or hard copy format**. The output device translates processed data from a machine coded form to a form that can be read by the end- user. The most common output devices that are in use are visual display units and printers.

Backing storage unit

Supplements the main storage by storing processed data examples include hard disk, flash disk, DVD and CDROM

Central processing unit (CPU)

It controls the hardware and carries out arithmetic and logic operations on user data. It acts as the “brain” of the computer. It consists of

- Main memory
- ALU
- Control unit

Main memory unit

The main function of main memory is to act as a buffer between the CPU and the rest of the components of the system. Types of computer memory are:

Random Access Memory (RAM)

- RAM is a location within the computer system or primary memory which is responsible for **storing data** on a temporary basis. Its contents can be overwritten.
- The information stored in RAM is typically loaded from the computer's hard disk, and includes data related to the operating system and certain applications.
- When the system is switched off, RAM loses all the stored information. The data remains stored and can be retained only when the system is running thus RAM is a **volatile memory**
- When the RAM gets full, the computer system is more likely to operate at **a slow speed**.
- RAM stores programs and data waiting to be processed, data being executed and processed data waiting to be communicated to the recipients.
- When many programs are running on the computer simultaneously, **the virtual memory** allows the computer to search in RAM for memory portions which haven't been utilized lately and copy them onto the hard drive. This action frees up RAM space and enables the system to load different programs.

Read Only Memory (ROM)

- This type of memory is active, regardless of whether the system is turned on or is switched off. It is a kind of permanent **non-volatile** storage memory.
- As the name 'read only' suggests, the contents in it cannot be changed or modified.
- It is an integrated circuit which is pre-programmed with important data that should necessarily be present for the computer to carry out its normal functionalities.
- ROM stores data and programs that are required for the initial activation of the computer system

Other types of memory

Cache memory

Cache is a kind of RAM which a computer system can access more responsively than it can in regular RAM. The processor looks up in the cache memory before searching in the central memory storage area to determine the information it requires.

This rules out the need for the system to search for information in larger and bigger memory storage areas, which in turn leads to a faster extraction of data.

Flash memory

This is a non-volatile kind of memory which is intended to contribute to portable storage and a convenient transfer of data from one computer to another.

The data in it can be **erased and re-programmed** as per the user's requirements. It only has a specific number of erase and write cycles that it can withstand, after which it creates a tendency to lose out on the stored information. Memory cards and USB flash drives are some modes of this type of memory storage.

Arithmetic & Logic unit (ALU)

Arithmetic logic unit is that part of the computer which performs the different logic operations and arithmetic calculations like addition, subtraction, multiplication and division. It is the high speed digital circuit that solves calculations and does comparisons.

Almost all the actions of the computer are done by it. It gets its data from a certain computer memory called processor register. After the data gets processed, its results get stored in output registers of the arithmetic logic unit.

Control unit

For each instruction the control unit causes the CPU to go through a sequence of control steps and in each control step the control unit issues a set of signals which cause the corresponding micro-operations to be executed.

The control unit is driven by the processor clock and the signals to be generated at a certain moment depend on the actual step to be executed, the condition and status flags of the processor, the actual instruction executed and the external signals received on the system bus (e.g. interrupt signals).